Unit 5 Quiz: Study Design

Name _____

Show your work.

5.1 - Samples, Studies, and Bias

1. **I name the population and the sample in a study, and I tell a sample survey, an observational study, and an experiment apart.**

   A music app has 60,000 users. To learn how users feel about a new design, the app sends a survey to 500 randomly chosen users.

   (a) What is the population?

   (b) What is the sample?

   Is each study a sample survey, an observational study, or an experiment? Explain.

Study	Type of study	Explain
(c) The music app study above		
(d) A researcher counts the birds that visit 20 yards with trees and 20 yards with no trees.		
(e) A teacher randomly chooses half of her students to read a story on paper. The others read it on a screen. All take the same quiz.		

2. **I name the type of bias in a sample or a question, and I decide if results can be generalized to a population.**

   Name the type of bias. Explain how it could change the results.

   (a) A TV show asks viewers to go to its website and vote: should the show move to a later time?

   (b) A survey asks, “Should the city waste money on a new soccer field?”

   (c) A gym emails a survey to 400 randomly chosen members. Only 90 members answer.

   (d) Lina asks the 25 students in her computer club how many hours a day they use their phones. Can she generalize her results to all students at her school? Explain.

5.2 - Sampling Methods

3. **I name and describe simple random, stratified, cluster, systematic, and convenience samples, and I tell stratified and cluster samples apart.**

   A library has 1,500 members. It wants to survey 60 of them about new hours. Name each sampling method.

How the library chooses 60 members	Method
(a) A computer randomly picks 60 member numbers.	
(b) A random member from the first 25, then every 25th member.	
(c) 20 randomly chosen children, 20 teens, and 20 adults.	
(d) The first 60 members who walk in on Saturday.	

(e) A city randomly selects 4 of its 40 apartment buildings. It surveys every person who lives in those 4 buildings. Which sampling method does the city use? Circle the letter.

A Simple random B Stratified C Cluster D Systematic

(f) Explain how a stratified sample of the people in the 40 buildings would be different from the city's sample.

4. **I use a random digit table to choose a sample, and I choose a good sampling method for a situation.**

   A coach has 40 players, numbered 01 to 40. Use this row of random digits to choose 4 players.

   34603 36904 52041 50165 61689 44176 49590 03997

   (a) Write the pairs you read. Then write the 4 players you choose.

   A school has 900 students. 300 ride the bus, and 600 do not. The principal wants to ask 90 students about a new bus schedule.

   (b) Describe a good sampling method. Name it, and explain why it is a good choice.

   (c) The principal asks the first 90 students who get off the buses. What is wrong with this method?

5.3 - Experiments

5. **I name the subjects, treatments, explanatory variable, and response variable in an experiment, and I explain why random assignment matters.**

   A track coach wants to know if a new warm-up helps runners run faster. The 24 runners on the team are randomly split into two groups of 12. One group does the new warm-up. The other group does the usual warm-up. Then each runner runs 400 meters, and the coach records the time.

(a) Subjects	
(b) Treatments	
(c) Explanatory variable	
(d) Response variable	

(e) Why should the coach split the runners into groups at random? Explain.

(f) Instead, the coach lets each runner choose a warm-up. Why is this a problem?

6. **I explain control groups, placebos, and blinding, and I describe completely randomized, randomized block, and matched pairs designs.**

   A company says its new “focus gum” helps students remember. 40 student volunteers are randomly split into two groups of 20. One group chews focus gum. The other chews plain gum that looks and tastes the same. Then all students take the same memory quiz, and a teacher grades it.

   (a) Which group is the control group?

   (b) The plain gum is a placebo. Why give the control group a placebo, and not nothing at all?

   (c) The teacher who grades the quizzes does not know which gum each student chewed. Why?

   (d) This experiment is a completely randomized design. Describe how to test the gum with a matched pairs design instead.

   (e) 16 of the 40 students are in 9th grade and 24 are in 12th grade. Older students may remember more. Describe how to test the gum with a randomized block design.

5.4 - What a Study Can Prove

7. **I decide if a study's results can be generalized to a population, and I decide if a study can show cause and effect.**

   For each study, can we conclude cause? Can we generalize to the population? Explain.

Study	Cause? Why?	Generalize? Why?
(a) Researchers randomly choose 600 teens in Missouri. Teens who spend more time outside sleep more.		
(b) 60 volunteers from a typing class are randomly assigned to Game A or Game B. The Game A group types faster.		

(c) A researcher randomly selects 200 students from a school district. She randomly assigns half to study with flash cards and half with a video. The flash card group scores higher, and the difference is statistically significant. Which conclusion is best supported?

A Studying with flash cards causes higher scores for all students in the United States.

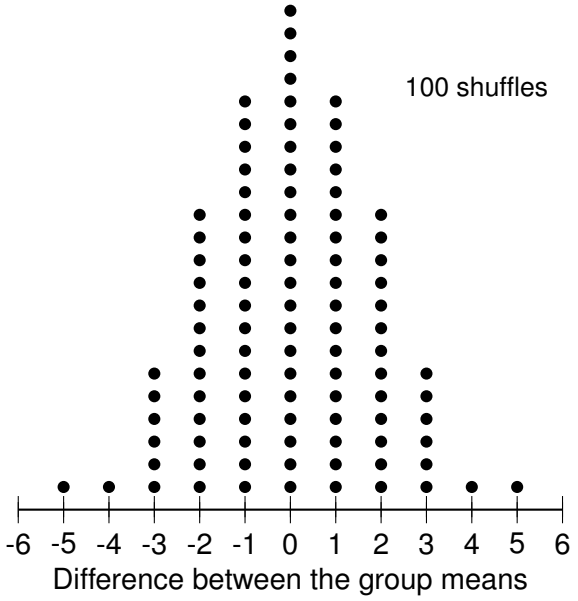
B Studying with flash cards causes higher scores for students in this school district.

C Flash cards are associated with higher scores, but the study cannot show cause.

D No conclusion can be made, because not every student in the district was studied.

8. **I use a simulation to decide if a result is statistically significant, and I find mistakes in conclusions about studies.**

A teacher tests a new vocabulary game. 30 students are randomly split into two groups of 15. The game group's mean is 4 points higher. Each dot is the difference between the group means in one random shuffle of the 30 scores, when the game does nothing.



(a) How many shuffles gave a difference of 4 or more?

(b) Is a 4-point difference statistically significant? Explain.

(c) Find the mistake. Suppose the game group had scored only 2 points higher. The teacher still says, "The game works."

(d) Find the mistake. A survey of 500 randomly chosen adults finds that dog owners walk more. A headline says, "Owning a dog makes people walk more."